

Vivek Kumar

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EDUCATION

University of Minnesota Twin Cities

Minneapolis, MN

Master of Science in Robotics

Sep 2024 – Present

Relevant Coursework: Machine learning, Computer Vision, Deep Learning, Intelligent Robotic Systems

National Institute of Technology Rourkela

Odisha, India

Bachelor of Technology in Mechanical Engineering

2019 – 2023

SKILLS

Programming & Frameworks: Python, C++, ROS 2, PyTorch, OpenCV, TensorRT, ONNX, Hugging Face

AI: Vision-Language Models (VLMs), Vision-Language-Action (VLAs), Diffusion Models, Imitation Learning, RL

Robotics & Hardware: SLAM, Path Planning, Nav2, Mapping, Sensor Fusion, Camera, LiDAR

Simulation & Tools: Isaac Sim, Gazebo, Rviz, MoveIt, Docker, SolidWorks, Linux

EXPERIENCE

Networking and AI Research Lab | Graduate Research Assistant

Minneapolis, MN

Autonomous Driving, Vision-Language-Action Models, Embodied AI, VLMs

Jun 2025 – Present

- Developing a sign-guided **Vision-Language-Action** framework for long-horizon **mapless navigation** that leverages semantic scene understanding and language reasoning to navigate toward user-specified destinations.
- Developed REACT2, a **reasoning model** using Qwen3-VL for **multi-robot supervision**, fine-tuned it on custom dataset containing **6000+** preference pairs covering long-tail autonomous mobile robots scenarios.
- Built a VLM model for proactive escalation in **Autonomous Driving** using Qwen3-VL and **NVIDIA Cosmos** as teacher model, fine-tuned on a **nuScenes** dataset with edge-case based sampling via **LoRA** and **DPO**, achieving **92.1%** accuracy.
- Developed an **ADAS** stack using YOLOv8m for **detection**, ByteTrack for **tracking**, and UFLD-v2 (ResNet-18) for **lane detection**, achieving **15 FPS** real-time inference via TensorRT/ONNX acceleration.
- Implemented a real-time **Bird's-Eye-View (BEV)** pipeline using **ROS 2**, utilizing Inverse Perspective Mapping (IPM) to provide teleoperators with a seamless surround-view, significantly reducing perspective distortion.
- Implemented and adapted a **real-time video prediction** model for autonomous driving, trained it on the **BDD100K** dataset and integrated it into a teleoperation pipeline to mitigate **100ms+** network latency through future-frame prediction.

GLOVEX Lab (UMN) | Graduate Researcher

Minneapolis, MN

Inverse Problems, Diffusion and Flow-matching models

April 2025 – Present

- Conducting research on **inverse** problems using generative priors like **diffusion** and **flow-matching** models, focusing on blind inpainting and zero-shot reconstruction from incomplete observations.
- Built ZS-PDP, a novel zero-shot **blind inpainting** framework, outperforming both **SOTA supervised** and **zero-shot** methods.

PUBLICATIONS

- Vivek Kumar**, et al. *VLM-Driven Proactive Escalation for Tele-operations in Autonomous Vehicles*. **CVPR Precognition Workshop 2026, Denver, CO**.
- Vivek Kumar**, et al. *Robots That Know When to Ask: VLM-Driven Reasoning for Human-on-Demand Multi-Robot Teleoperation*. (Under Review) [Demo](#)
- W Zhang, **Vivek Kumar**, et al. *Zero-Shot Blind Inpainting with Pretrained Diffusion Priors*. **NeurIPS 2026** (Under Review)

PROJECTS

Full-Stack Autonomous Navigation System for AgileX Hunter 2.0 | [Demo](#)

ROS 2, C++, Python

- Developed a production-ready modular **autonomy stack** on real hardware using ROS 2 by integrating **SLAM**, LiDAR **localization**, D*+ **path planning**, and PID-based control for autonomous indoor navigation in pre-mapped environments.
- Built the **collision avoidance** pipeline using occupancy grid maps and **octomap** to support safe autonomous navigation in human-shared spaces and dynamic environments.

Multimodal Diffusion Policy for Autonomous Driving

Python, PyTorch

- Trained a lightweight Vision-Language-Action (**VLA**) **diffusion policy** on a custom driving dataset, utilizing a **CNN-Transformer** architecture to predict autonomous trajectories from natural language instructions.

On-Demand Mobile EV Charging Robot (WEZZON)

ROS 2, Agile-X

- Built an autonomous mobile EV charging robot to mitigate fixed infrastructure barriers and validated it through real-world parking lot testing using a battery-integrated AgileX Hunter platform.

Robot Manipulation via Action Chunking Transformers in ManiSkill

Python, PyTorch, ManiSkill

- Built an end-to-end **imitation learning** pipeline using Action Chunking Transformer (ACT) architecture to train robot arm manipulation policy on RGBD observations, achieving **95%+** success rate on PickCube task with 30K training iterations.